Date

5.9 (1) 设原式 $= 246_3$

(2) 原式 $= 6_1^3 - 3\cancel{6_2 6_3} 3 6_1 6_2$

2、$p = \cancel{x_1 x_2} 6_2 \quad r = -6_3, \quad q = -6_1$

$\therefore (X - x_1^2)(X - x_2^2)(X - x_3^2) = X^3 - (x_1^2 + x_2^2 + x_3^2)X^2 + (x_1^2 x_2^2 + x_2^2 x_3^2 + x_1^2 x_3^2)X - (x_1 x_2 x_3)^2$

$= X^3 - (6_1^2 - 26_2)X^2 + (6_2^2 - 26_1 6_3)X - 6_3^2$

$= X^3 - (q^2 - 2p)X^2 + (p^2 - 2rq)X - r^2$

3、设三根为 x_1, x_2, x_3，则题意为

$(x_1^2 + x_2^2 - x_3^2)(x_1^2 + x_3^2 - x_2^2)(x_2^2 + x_3^2 - x_1^2) = 0 \Leftrightarrow p^4(p^2 - 2q) = 2(p^3 - 2pq + 2r)^2$

$\therefore$ 左侧 $(x_1^2 + x_2^2 - x_3^2)(x_1^2 + x_3^2 - x_2^2)(x_2^2 + x_3^2 - x_1^2) = -6_1^6 + x_1 6_1^4 6_2 + x_2 6_1^3 6_3 + x_3 6_2^2 6_3^2 + x_4 6_1 6_2 6_3$

$+ x_5 6_2^3 + x_6 x_3^2 = 0$

代入 x_1, x_2, x_3 的一些特殊值以及 $p = 6_2 \quad r = -6_3, \quad q = -6_1$

得 $p^6 - 2qp^4 - 2(p^3 - 2pq + 2r)^2 = 0$

$\therefore p^4(p^2 - 2q) = 2(p^3 - 2pq + 2r)$

$\therefore$ 得证

4、$n=1$ 时 $S_4 = 6_1^4$

$n=2$ 时 $S_4 = S_3 6_1 - S_2 6_2 \quad S_3 = S_2 6_1 - S_1 6_2 \quad S_2 = S_1 6_1 - 26_2 = 6_1^2 - 26_2$

$\therefore S_3 = 6_1^3 - 36_1 6_2 \quad \therefore S_4 = 6_1^3 - 36_1^2 6_2 - (6_1^2 - 26_2)6_2$

$= 6_1^4 - 46_1^4 6_2 + 2 6_2^2$

$n=3$ 时 $S_4 = \cancel{6} S_3 6_1 - S_2 6_2 + S_1 6_3 \quad S_3 = 6_1^3 - 36_1 6_2 + 36_3 \quad S_2 = 6_1^2 - 26_2 \quad S_1 = 6_1$

$\therefore S_4 = 6_1^4 - 36_1^2 6_2 + 36_3 6_1 - 6_1^2 6_2 + 2 6_2^2 + 6_1 6_3$

$= 6_1^4 - \cancel{4} 6_1^2 6_2 + 2 6_2^2 + 4 6_3 6_1$

$n \geq 4$ 时 $S_4 = \cancel{6} S_3 6_1 - S_2 6_2 + S_1 6_3 \cancel{+ S_0 6_4}$

$\therefore S_4 = 6_1^4 - 4 6_1^2 6_2 + 2 6_2^2 + 4 6_1 6_3 - 4 6_4$

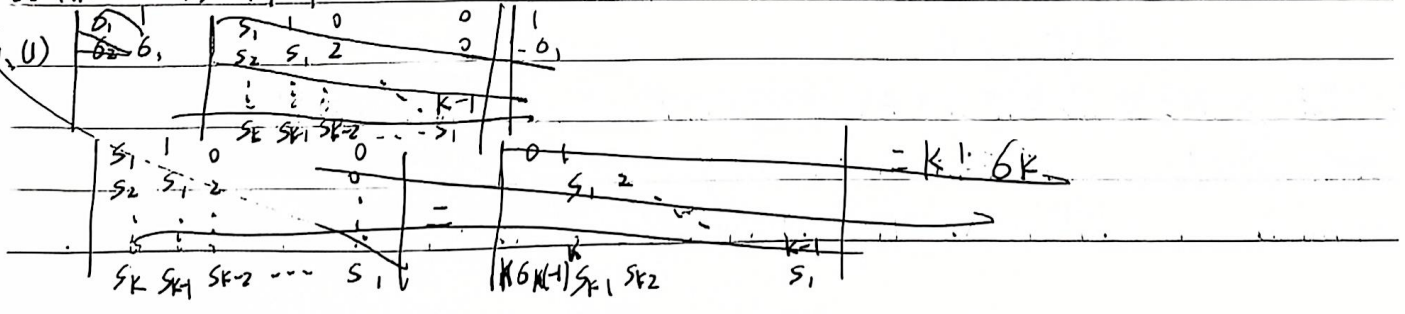
5、$S_1 = S_2 = S_3 = S_4 = 4 \quad n = 4$

$\therefore 6_1 = 4 \quad 6_2 = 6 \quad 6_3 = 4 \quad 6_4 = 1$

$\therefore x_1, x_2, x_3, x_4$ 是 $(x-1)^4 = x^4 - 4x^3 + 6x^2 - 4x + 1 = (x-1)^4 = 0$ 的四个根

$\therefore x_1 = x_2 = x_3 = x_4 = 1$

6、(1)



~~(2) 同(1)~~

6.(1) 第2列加上第1列的 -6_1 倍，第3列的 6_2 倍 …… 第K列的 $(-1)^{K-1}6_K$ 倍，得到

$$\frac{1}{K!}\begin{vmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & S_2 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & S_K & S_{K-1} & \cdots & S_1 \end{vmatrix}$$

以第1列展开，得 原式 $= 6_K\ K!\ X_{\overline{K1}} = 6_K$

(2) 同(1)

5.10 (1)
$$\begin{vmatrix} 1 & 3 & -1 & 4 \\ 1 & 3 & -1 & 4 \\ 1 & 3 & -1 & 4 \\ 1 & -2 & -1 & \\ & 1 & -2 & -1 \\ & & 1 & -2 & -1 \end{vmatrix} - \begin{vmatrix} 1 & 3 & -1 & 4 \\ & 1 & 3 & -1 & 4 \\ -5 & 0 & -4 & \\ 1 & -2 & -1 & \\ & 1 & -2 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 3 & -1 & 4 \\ & 1 & 3 & -1 & 4 \\ -5 & 0 & -4 & \\ 1 & -2 & -1 & \\ & 1 & -2 & -1 \end{vmatrix} = -\begin{vmatrix} 1 & 3 & -1 & 4 \\ & 1 & 7 & -9 \\ -5 & 0 & 4 & \\ 1 & -2 & -1 & \\ & 1 & -2 & -1 \end{vmatrix}$$

(1)
$$\begin{vmatrix} 1 & 3 & -1 & 4 \\ 1 & 3 & -1 & 4 \\ 1 & -2 & -1 & \\ & 1 & -2 & -1 \\ & & 1 & -2 & -1 \end{vmatrix} = -\begin{vmatrix} 1 & 3 & -1 & 4 \\ 5 & 0 & 4 & \\ -1 & 2 & 1 & \\ & -1 & 2 & 1 \end{vmatrix} = -\begin{vmatrix} 1 & 7 & 9 \\ 5 & 0 & 4 \\ -1 & 2 & 1 \end{vmatrix} = 161$$

(2)
$$\begin{vmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & & & \\ & 1 & 1 & 1 & & \\ & & 1 & 1 & 1 & \\ & & & 1 & 1 & 1 \end{vmatrix} = 1 - \begin{vmatrix} 0 & 1 & & & \\ 1 & 1 & 1 & & \\ & 1 & 1 & 1 & \\ & & 1 & 1 & 1 \\ & & & 1 & 1 & 0 \\ & & & & 1 & 1 \end{vmatrix} = 1$$

2. (2)
$$\begin{vmatrix} 1 & 0 & p & 9 & \\ & 1 & 0 & p & 9 \\ 3 & 0 & p & & \\ & 3 & 0 & p & \\ & & 3 & 0 & p \end{vmatrix} = \begin{vmatrix} -2p & -39 & 0 \\ 0 & -2p & -39 \\ 3 & 0 & p \end{vmatrix} = -4p^3 - 27q^2$$

3. (1) 对 $R(f,g)$ 的对应的行列式进行 mn 次初等行变化（交换）可得到 $R(g,f)$

$\therefore R(f,g) = (-1)^{m\times n} R(g,f)$

(2) 前 m 行提取 a，后 n 行提取 b，得 $R(af,bg) = a^m b^n R(f,g)$

4. 设 $f(x) = a f_1(x)$ $g_1(x) = b h_1(x)$ $g_2(x) = c h_2(x)$，$f(x), g_1(x), g_2(x)$ 分别为 t_1, t_2, t_3 次

$\therefore R(f,g_1g_2) = a^{t_2+t_3} b^{t_1} c^{t_1} R(f_1, h_1h_2)$

设 $f_1(x)$ 根为 $x_1 \cdots x_t$，h_1 为 $y_1 \cdots y_{t_2}$ h_2 为 $z_1 \cdots z_{t_3}$

$$R(f,g_1g_2) = a^{t_2+t_3} b^{t_1} c^{t_1} \prod_{i=1}^{t_1} f(x_i) \left(\prod_{j=1}^{t_2}(x_i - y_j) \prod_{j=1}^{t_3}(x_i - z_j)\right)$$

$$= a^{t_2} b^{t_1} a^{t_3} c^{t_1} R(f_1, h_1) R(f_1, h_2)$$

$$= R(f,g_1) R(f,g_2)$$

5. 已知 $f(x)$ 的虚根成对存在且本共轭也为 $f(x)$ 的根，设 x_1, x_2, x_3, x_4 为 $f(x)$ 的根，其中 x_3, x_4 为实根，x_1, x_2 为虚根且 $x_1 \neq \bar{x}_2$ $x_1 \neq x_2$

对 x_1, x_3 $(x_3 - x_1)^2 (x_3 - \bar{x}_1)^2 = [(x_3 - x_1)(x_3 + \bar{x}_1)]^2 \geq 0$

CS 扫描全能王
3亿人都在用的扫描App

Date

对 X_1, X_2 $(X_1-X_2)^2(X_1-\bar{X}_2)^2(\bar{X}_1-X_2)^2(\bar{X}_1-\bar{X}_2)^2=[(X_1-X_2)(X_1-\bar{X}_2)(\bar{X}_1-X_2)(\bar{X}_1-\bar{X}_2)]^2>0$

对 X_3, X_4 $(X_3-X_4)^2>0$

∴当且仅当仅对一对共轭虚数 $(X_1-\bar{X}_1)^2<0$

∴$\Delta(f)>0$，则虚根对数为偶数 $\Delta(f)<0$，则虚根对数为奇数

6、设 $f(x)=a_0x^n+\cdots+a_n$

$f(x)$的根为 $X_1\cdots X_n$

∴ $X_1X_2\cdots X_n=\frac{a_n}{a_0}(-1)^n$

∴ $\Delta(f(x))=a_0^{2n-2}\prod_{1\le i\le j\le n}(X_i-X_j)^2$

$\Delta(f(x^2))=a_0^{4n-2}\prod_{1\le i<j\le n}(\sqrt{X_i}-\sqrt{X_j})^2(\sqrt{X_i}+\sqrt{X_j})^2(-\sqrt{X_i}+\sqrt{X_j})^2\prod_{1\le i\le n}(2\sqrt{X_i})^2$

$=a_0^2\Delta^2(f(x))\,4^n(-1)^n\frac{a_n}{a_0}$

$=a_0a_n(-4)^n\Delta^2(f(x))$

7、设 $f(x)=t^3+2t-3-x$ $g(x)=t^2-t+1-y$

∵ $f(x)$与$g(x)$有共公共根，则

$$R(f,g)=\begin{vmatrix} 1 & 0 & 2 & -3-x & & \\ & 1 & 0 & 2 & -3-x & \\ 1 & -1 & 1-y & & & \\ & 1 & -1 & 1-y & & \\ & & 1 & -1 & 1-y & \end{vmatrix}=\begin{vmatrix} 1 & 0 & 2 & -3-x & \\ -1 & 1-y & 3+x & & \\ 1 & -1 & 1-y & & \\ & 1 & -1 & 1-y & \end{vmatrix}$$ 正

$=y^3-x^2+3xy+y^2-6x+10y+12=0$